CS341 · COMPUTER GRAPHICS

Introduction to Computer Graphics

Midterm summary

Muharib

What's inside

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01 What is computer graphics?

The official definition on the slide (asked word for word in the midterm):

> Computer graphics is commonly understood to mean the **creation, storage, and manipulation of models and images**.

Put more simply: generating 2D images of a 3D world represented in a computer.

> Computer graphics: generating 2D images of a 3D world represented in a computer.
> The term "computer graphics" was coined in **1960** by **William Fetter**.

Models come from physical, mathematical, artistic and biological fields, and even from conceptual (abstract) structures.

02 The three main tasks

Task	Meaning
Modeling	Creating and representing the geometry of objects in the 3D world.
Rendering	Generating 2D images of the objects.
Animation	Describing how objects change in time.

Asked before

> Which of the following is NOT a main task for computer graphics? Modeling / Rendering / Animation / Data encryption → **Data encryption**

03 Goals of computer graphics

- Synthetic images indistinguishable from reality.
- Practical, scientifically sound, in real time.

Computer graphics goals: **synthetic images indistinguishable from reality**; practical, scientifically sound, in real time.

04 Interactive computer graphics

The user controls the contents, structure and appearance of objects and sees the result immediately (rapid visual feedback). The first truly interactive graphics system was Sketchpad (MIT, 1963).

Component	Examples
Input	Mouse, tablet and stylus, force feedback device, scanner, live video streams
Processing (and storage)	Processing and storage
Display / output	Screen, paper-based printer, video recorder

05 Applications

- **Computer Aided Design (CAD)**
- **Medical Visualization**
- Entertainment: games and cinema
- Everyday use: user interfaces and large-screen interaction
- **Simulation**
- **Education**
- **Training**
- **Scientific Visualization**

Repeats in the midterm

"Computer graphics applications are used only in the entertainment industry." → **False**
"One of the applications of computer graphics: CAD / Medical visualization / Entertainment / All of the above" → **All of the above**

06 What the course covers

We cover

- Graphics programming algorithms
- Graphics data structures
- Color and human vision
- Applied geometry and modeling
- Graphics API

It is NOT

- Paint and imaging packages (Photoshop)
- CAD packages (AutoCAD)
- Rendering, modeling, animation packages
- Graphics modeling and languages (RenderMan)
- Game programming (GPU)
- User interface design
- Image processing